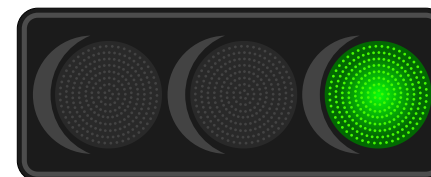


SPACE ROBOTICS

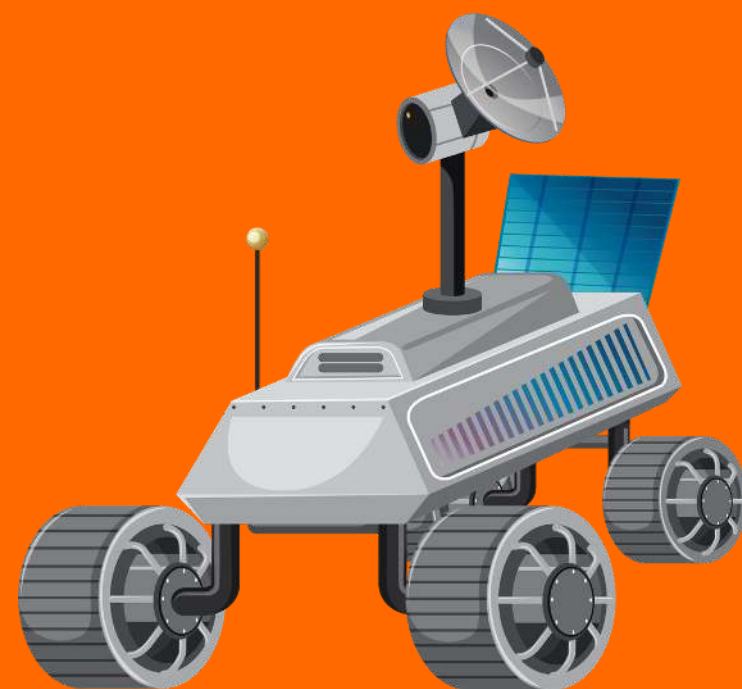
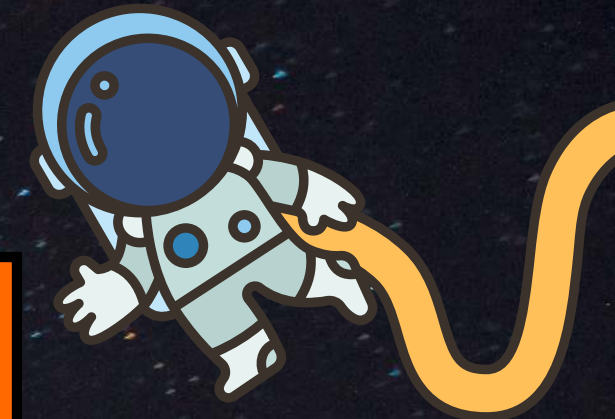
Moon Base



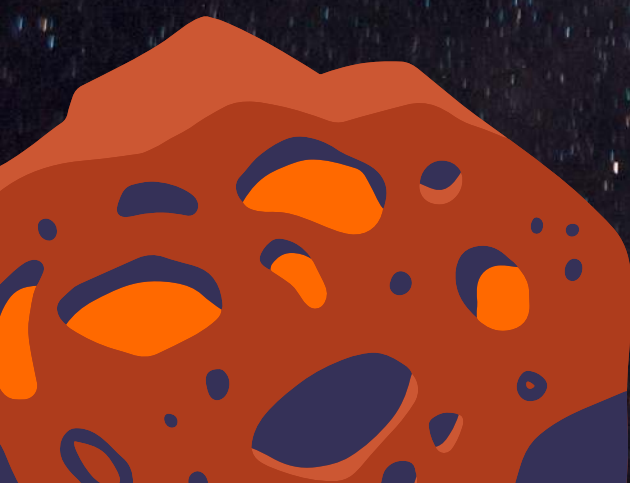
Level: Easy



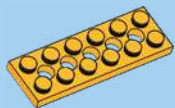
LITTLE
SCIENTISTS
ACADEMY



BUILD A ROVER ROBOT

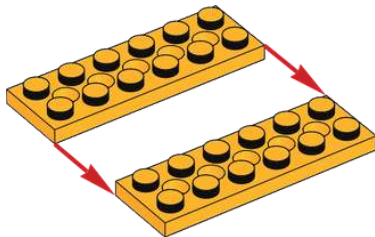


By: Scilla Roncallo



2x

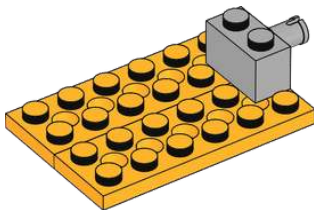
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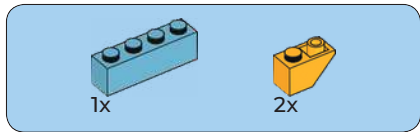




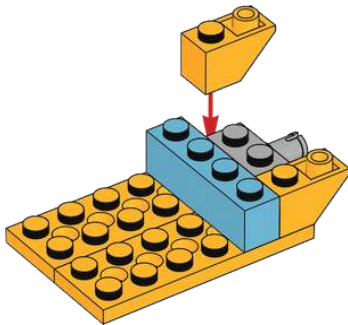
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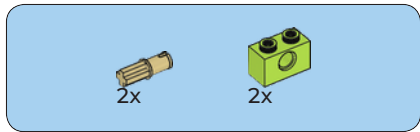
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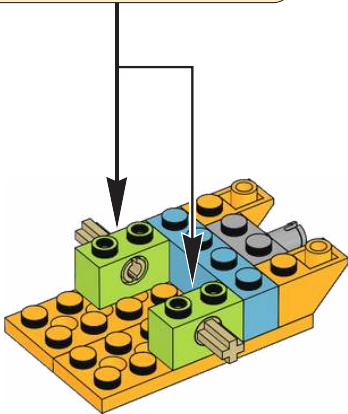
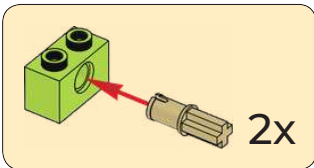


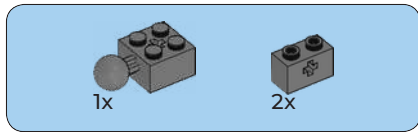
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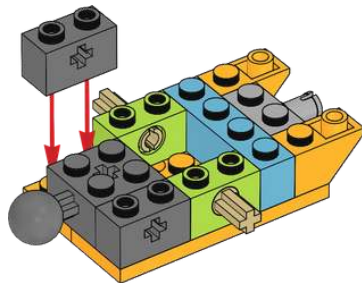


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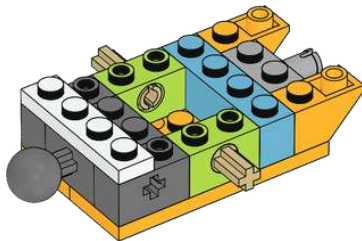
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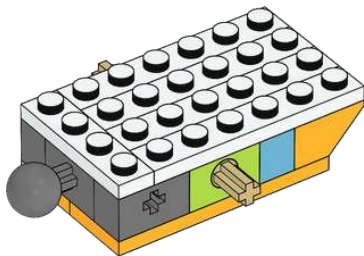
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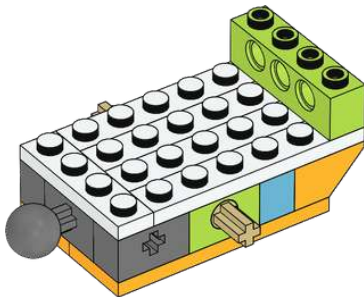
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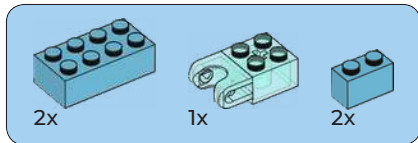




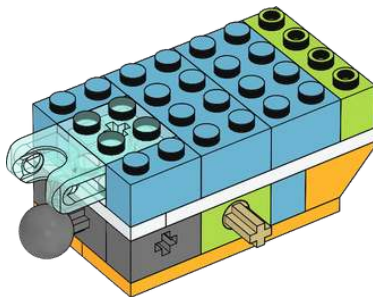
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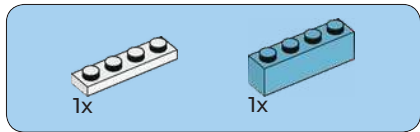
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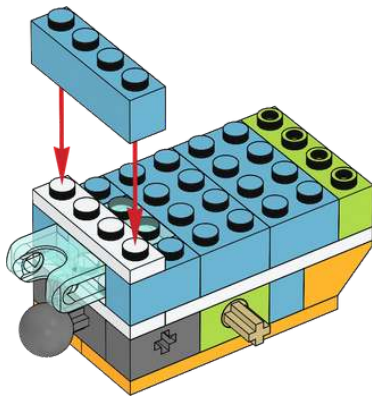


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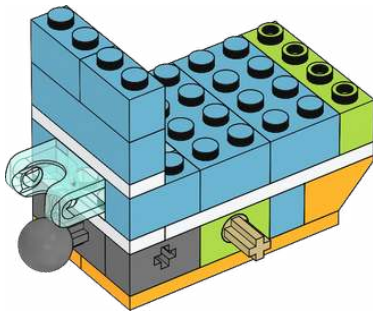
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2x

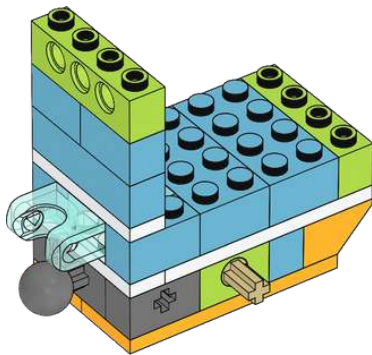
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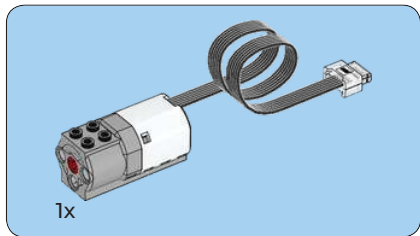




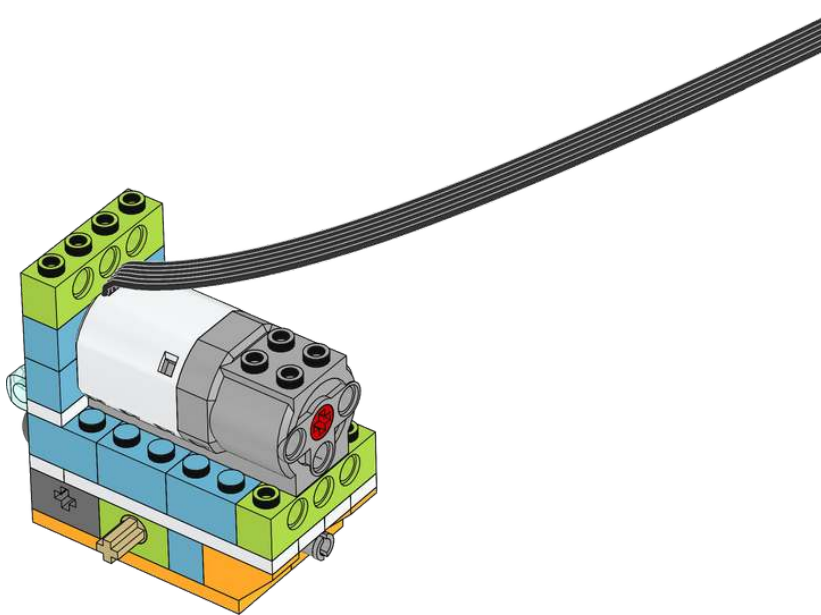
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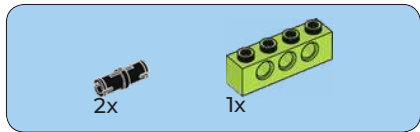
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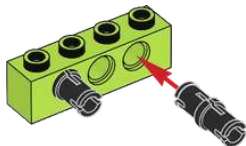


13





14



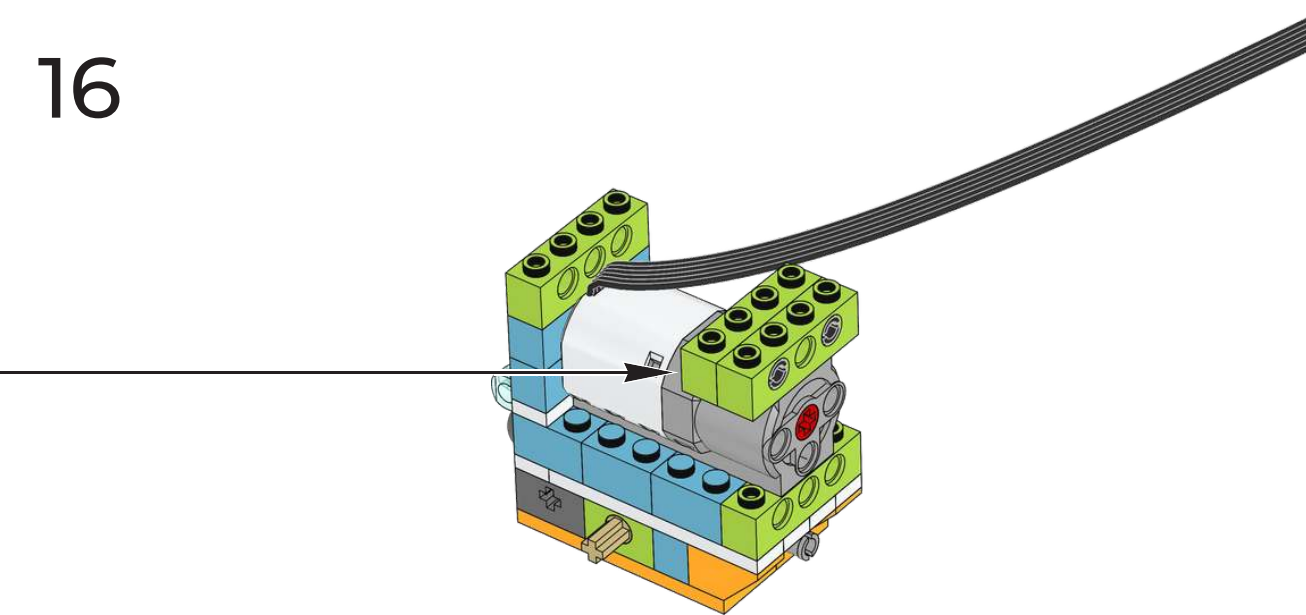


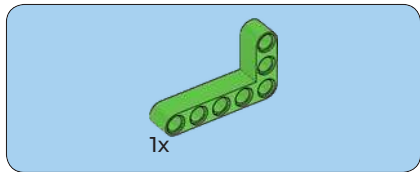
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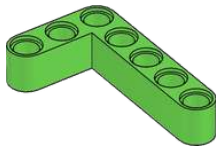


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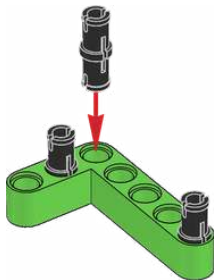
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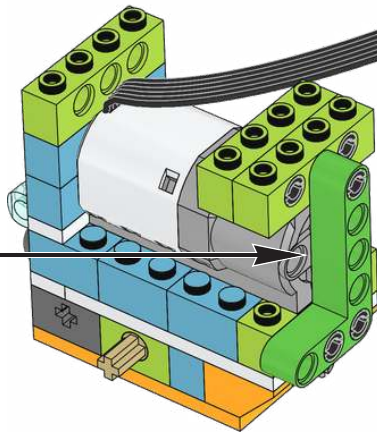


3x

18



19



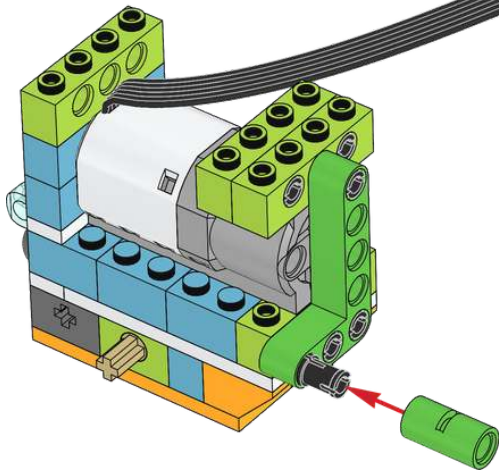


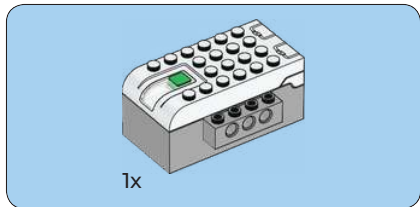
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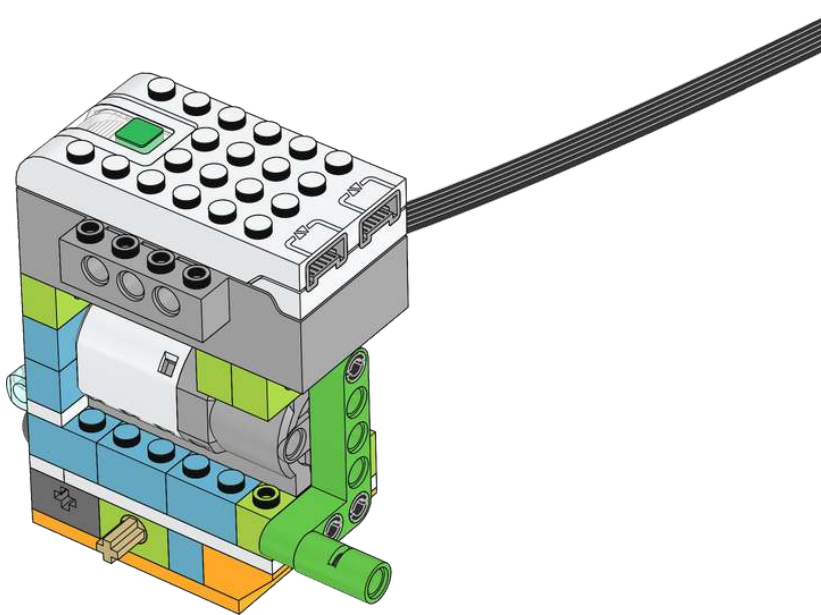
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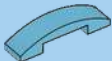
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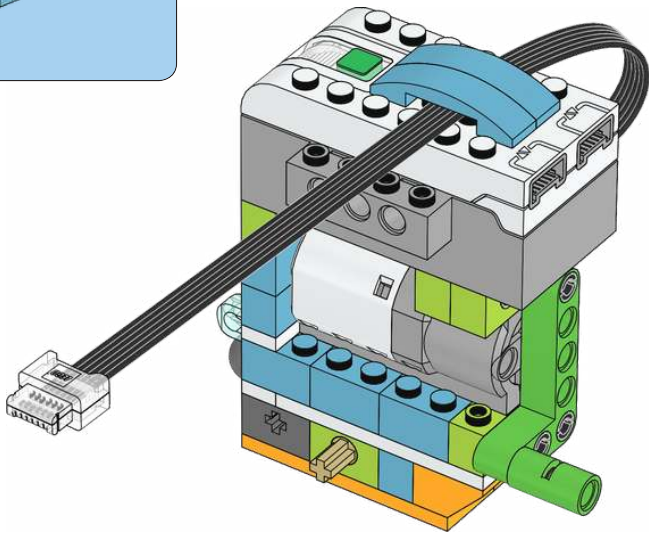
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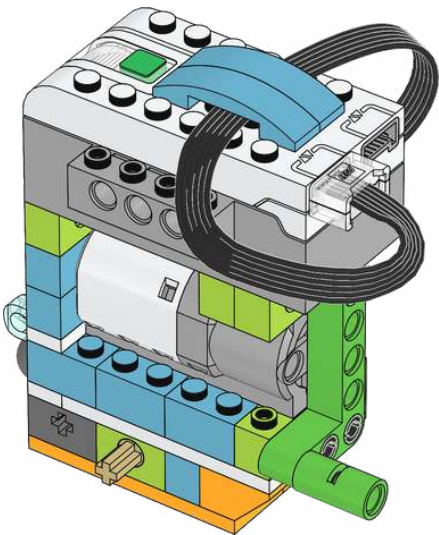


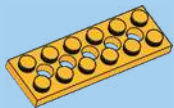
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22



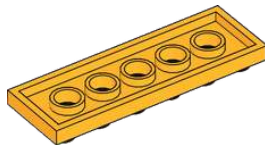
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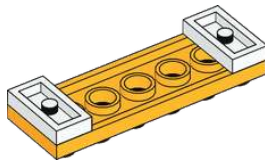
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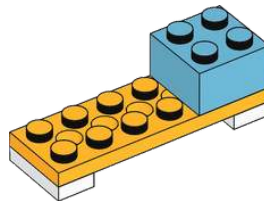
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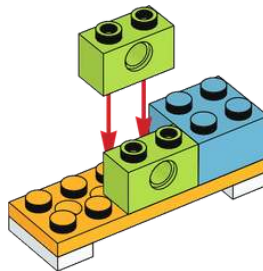
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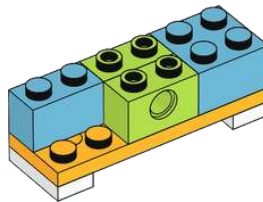
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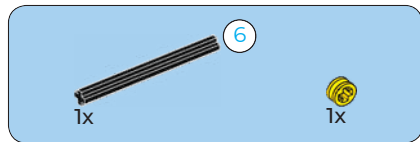




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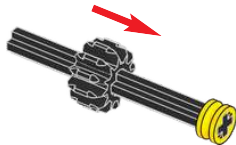
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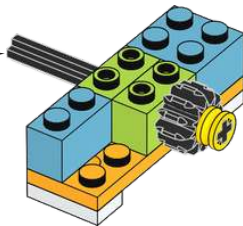


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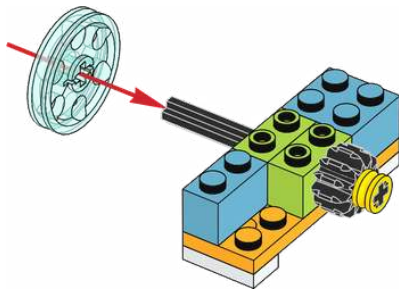
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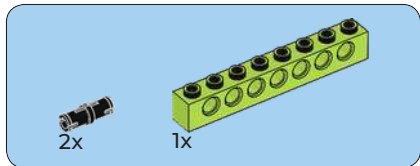




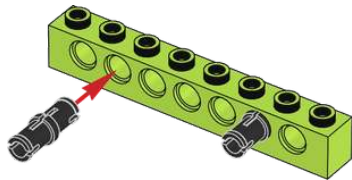
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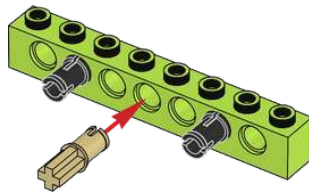


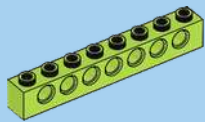
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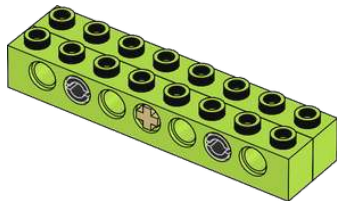
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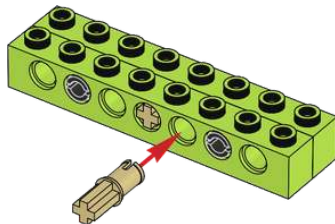
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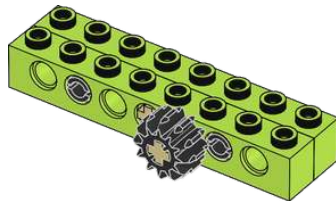
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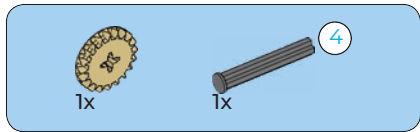




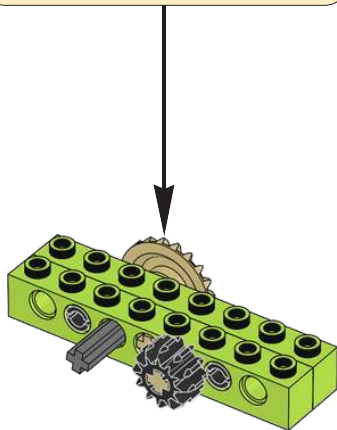
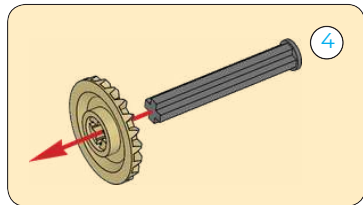
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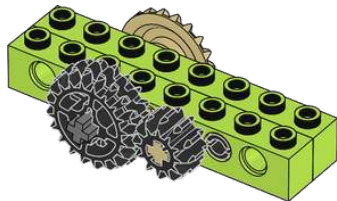
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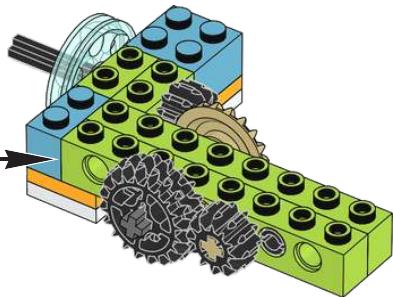


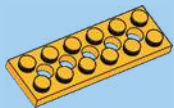
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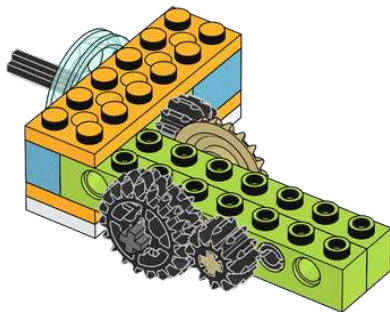
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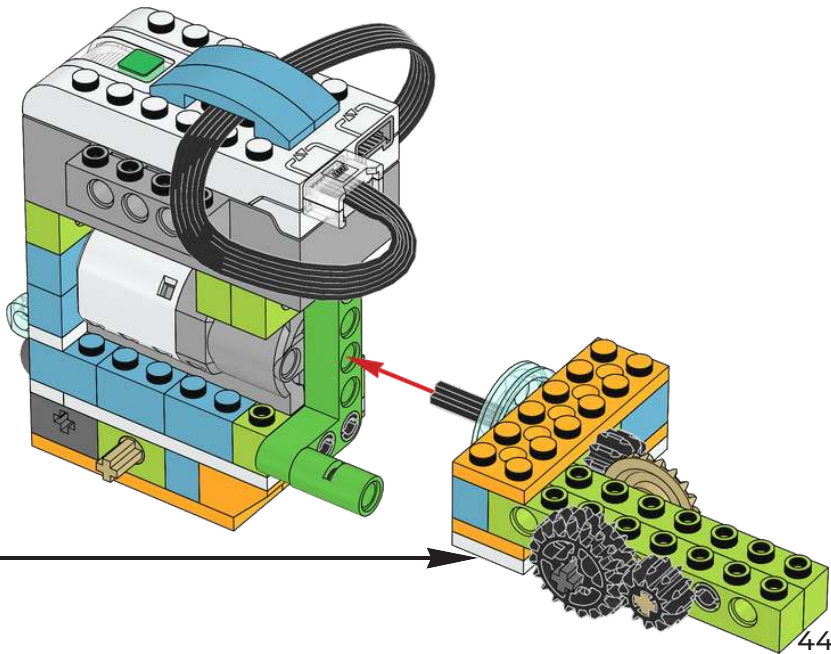


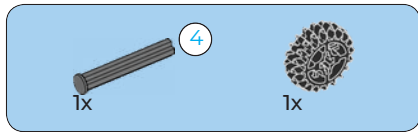
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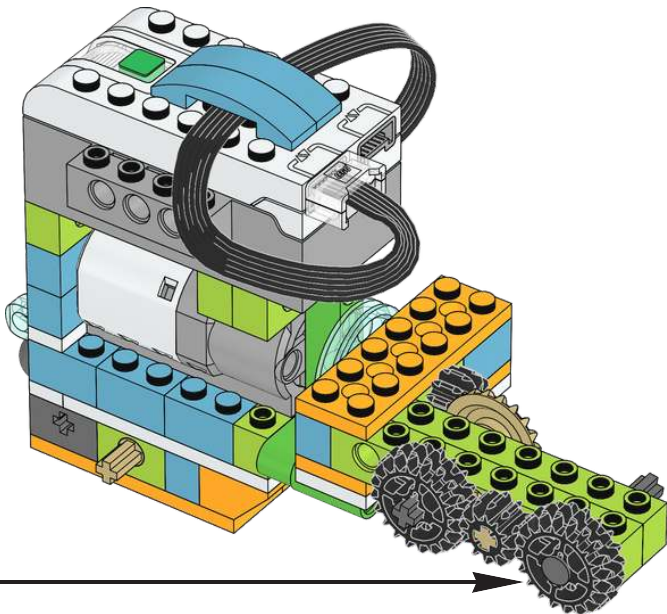
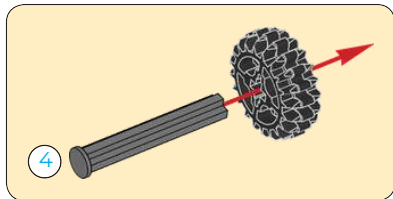


42





43



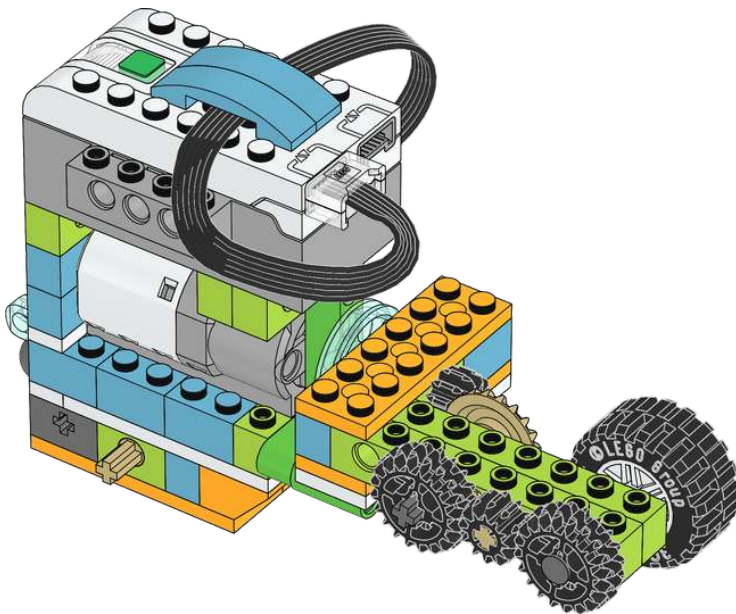


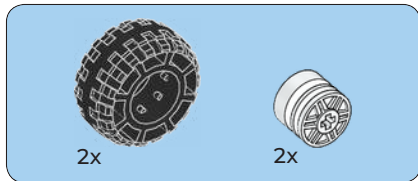
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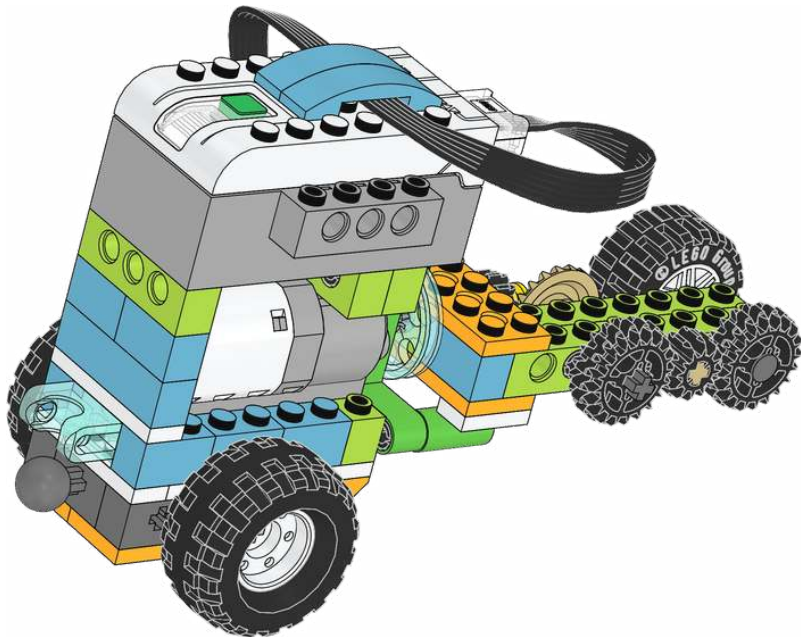
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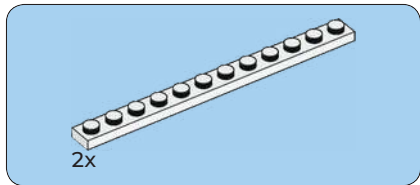
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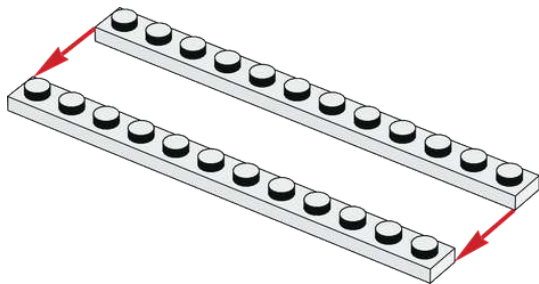


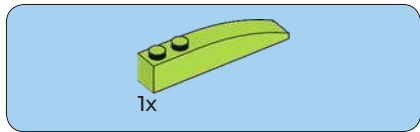
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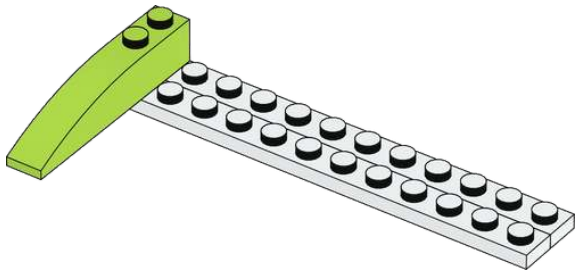


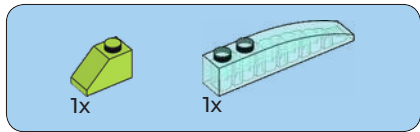
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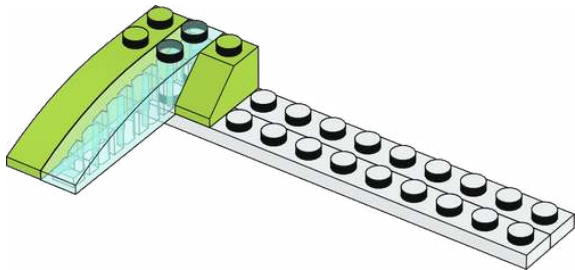


47





48



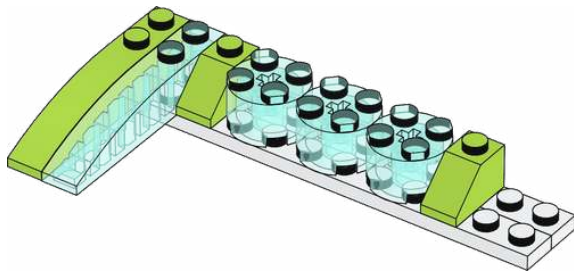


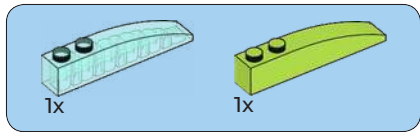
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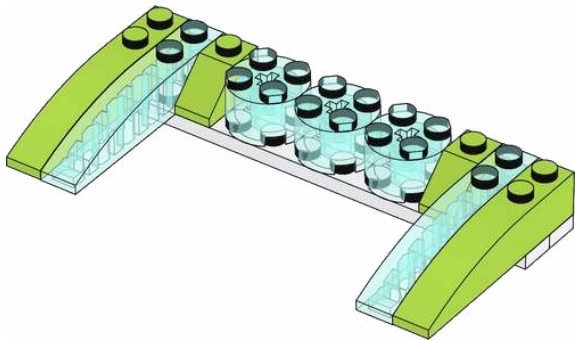
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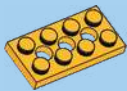
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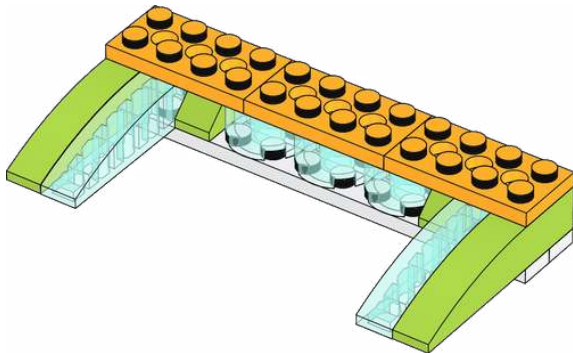
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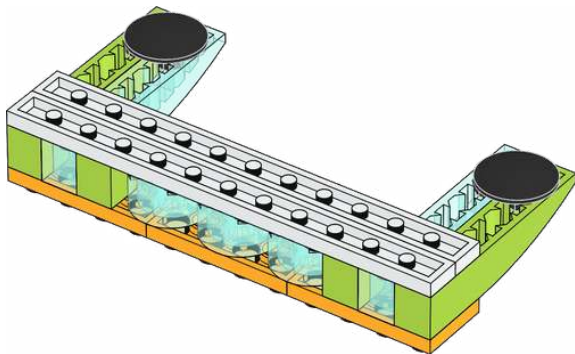
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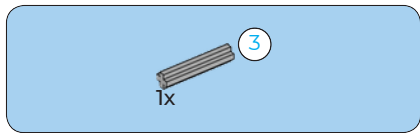




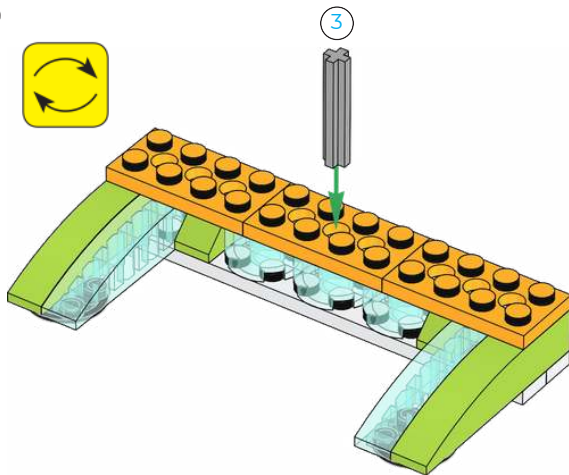
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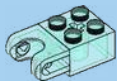
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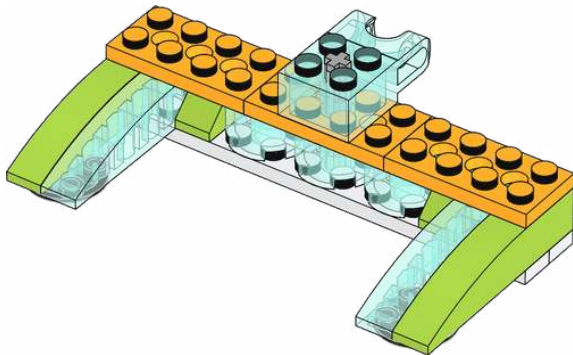
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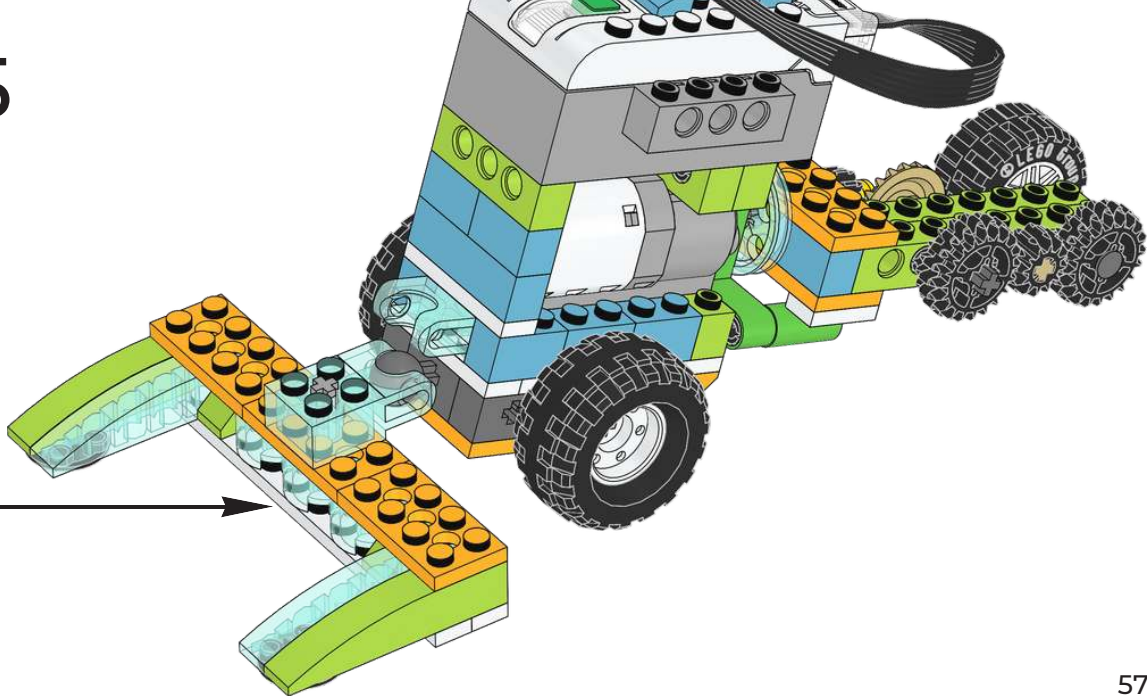


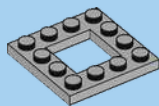
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54



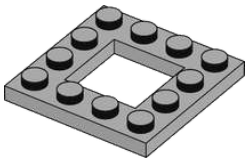
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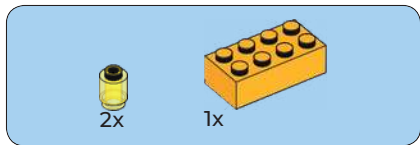




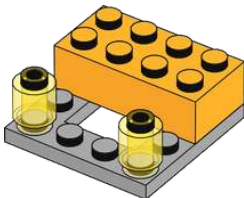
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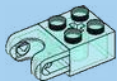
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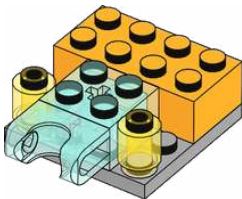
57





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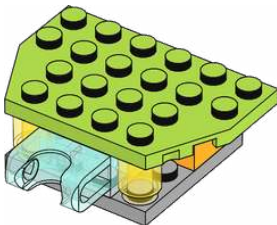
58





1x

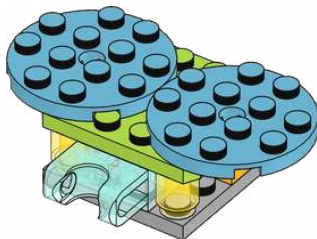
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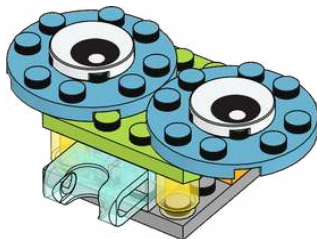
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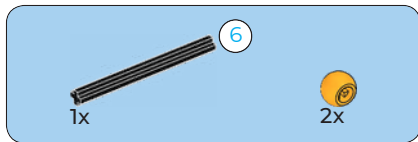
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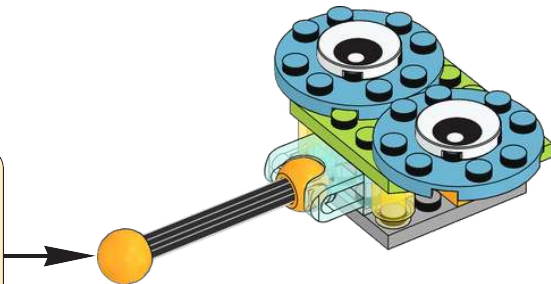
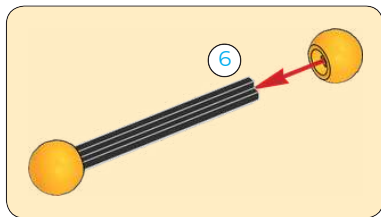


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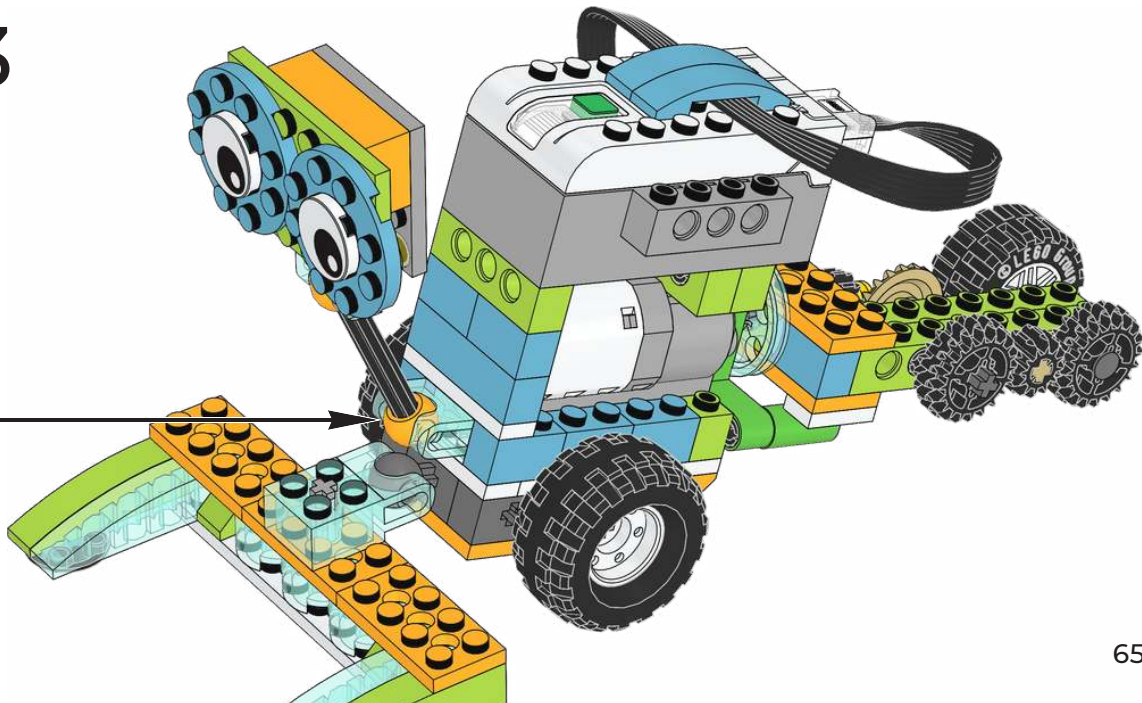


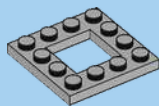


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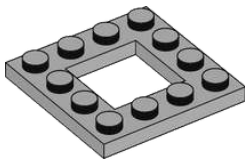
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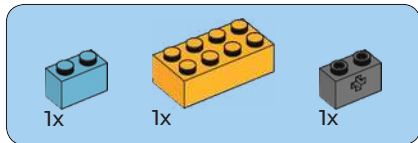




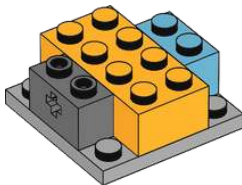
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64





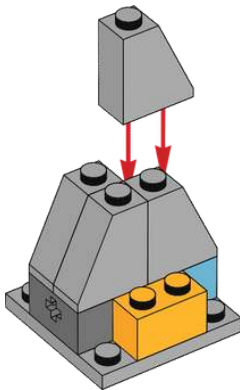
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4x

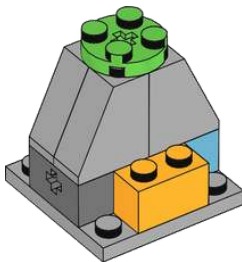
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1x

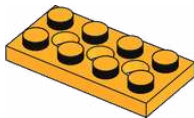
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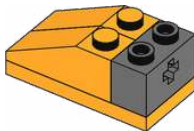
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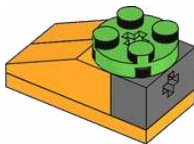
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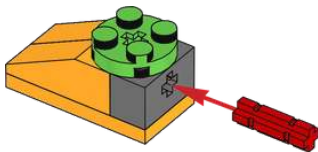
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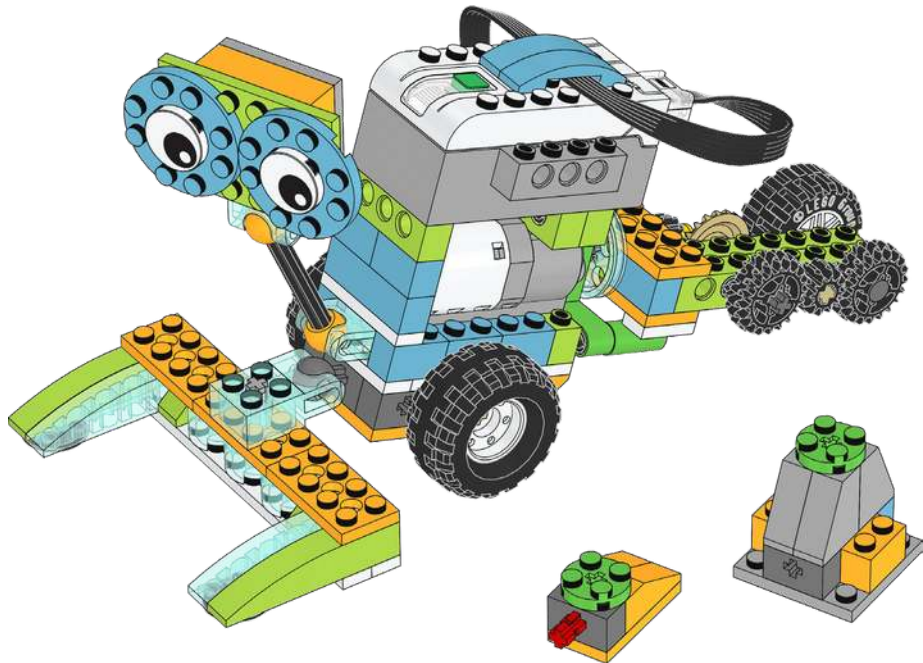




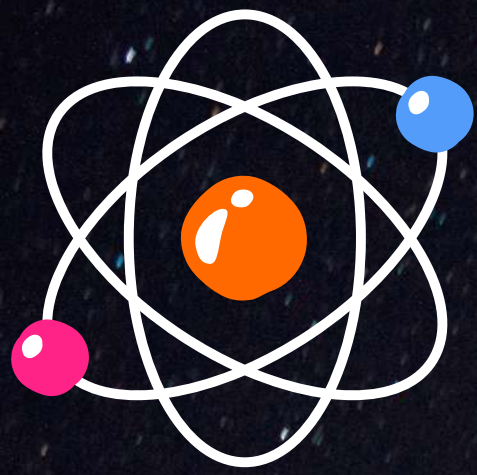
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72

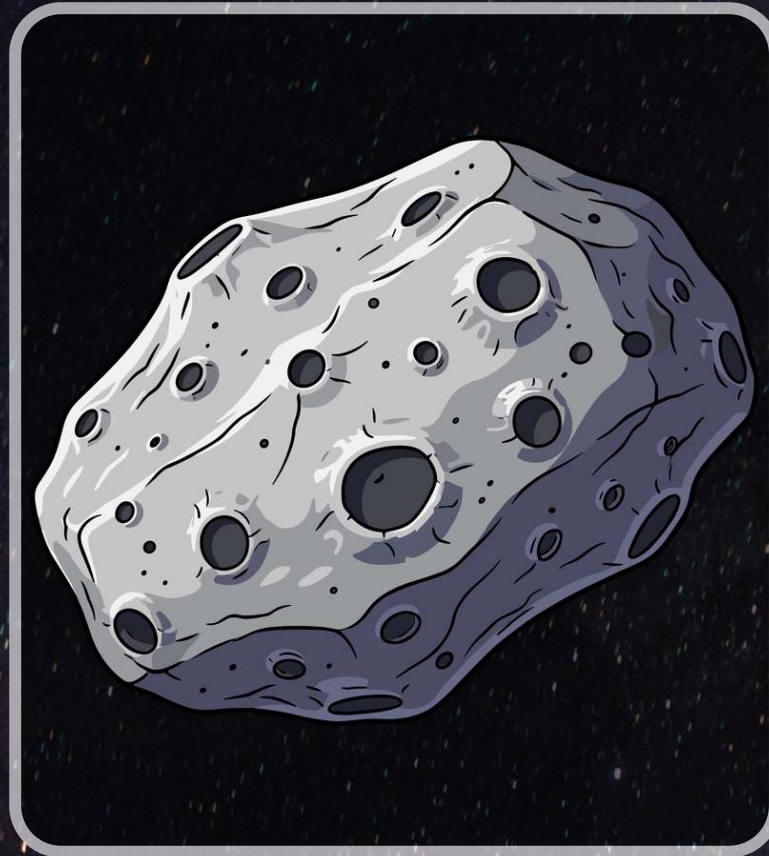
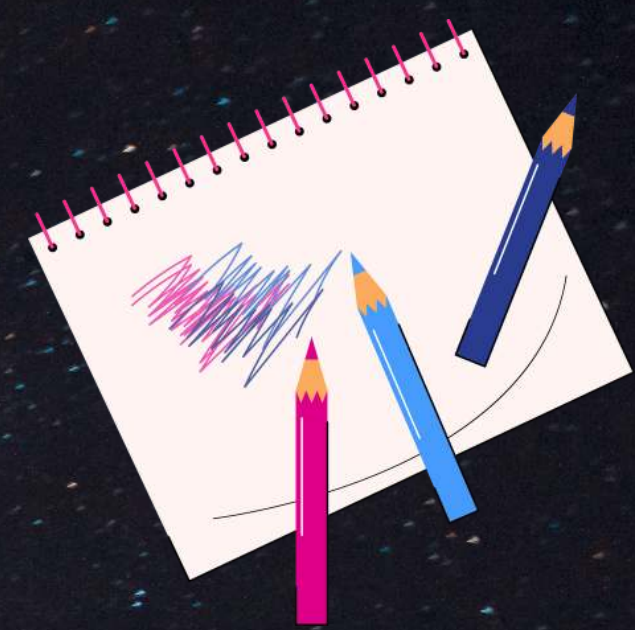


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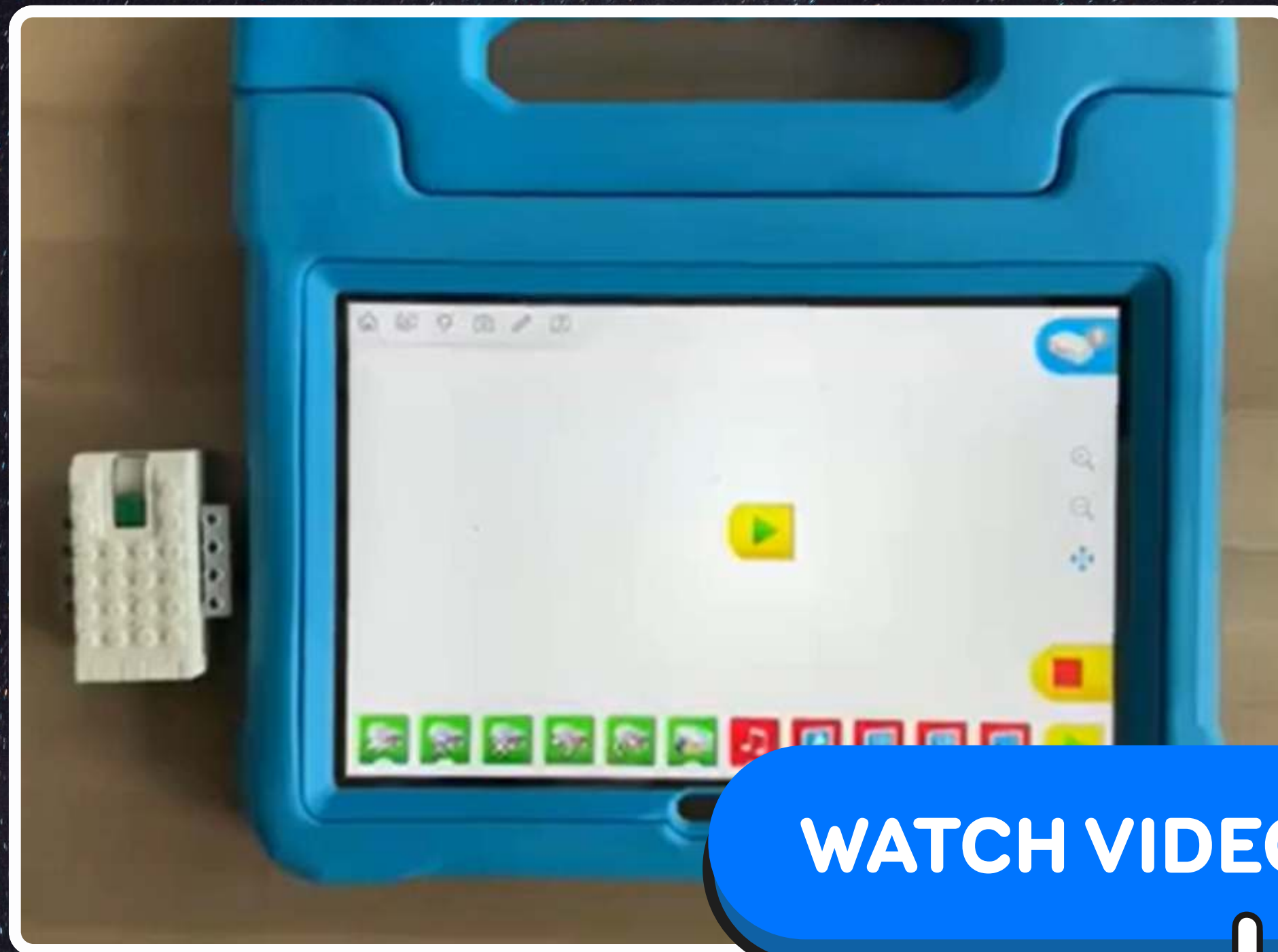
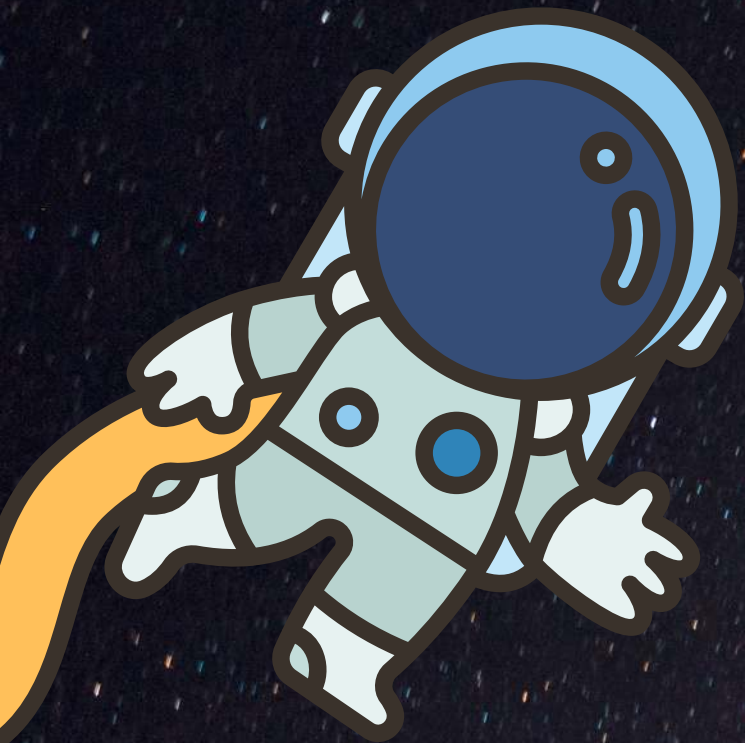
Challenge 2

CRATERS AND OSTACLES



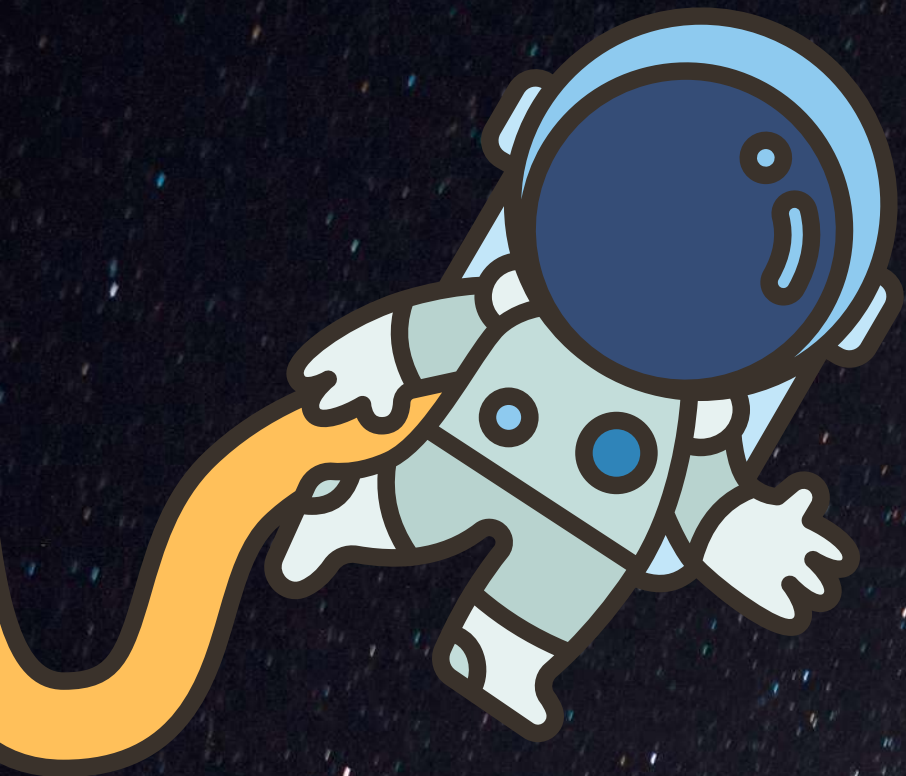
Create stones for your robot to collect
Erstelle Steine, die dein Roboter einsammeln kann

CONNECT TO WEDO APP



WATCH VIDEO

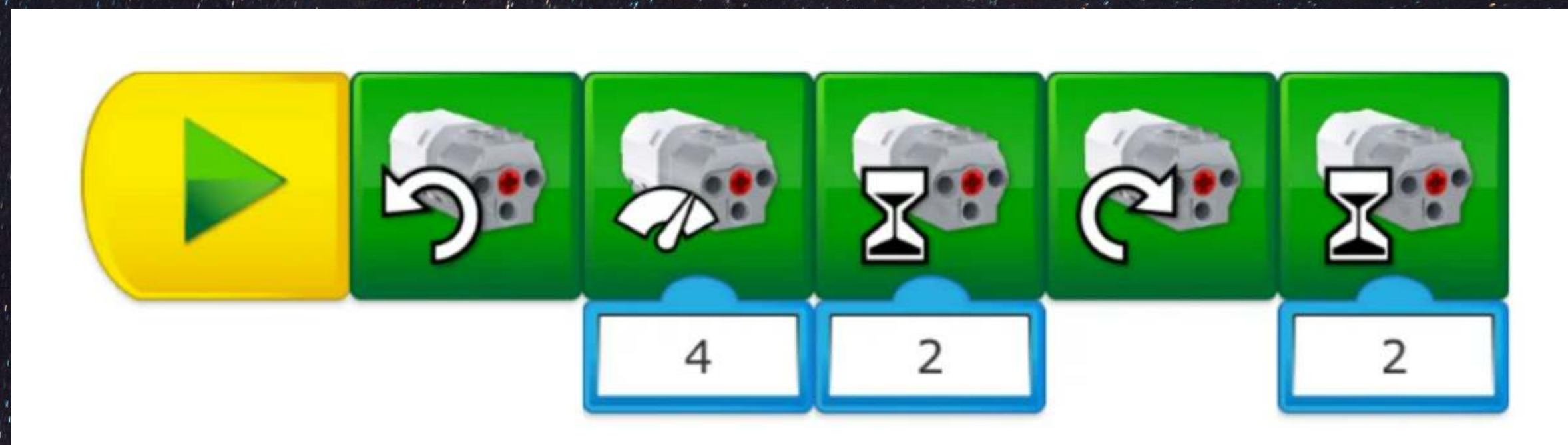




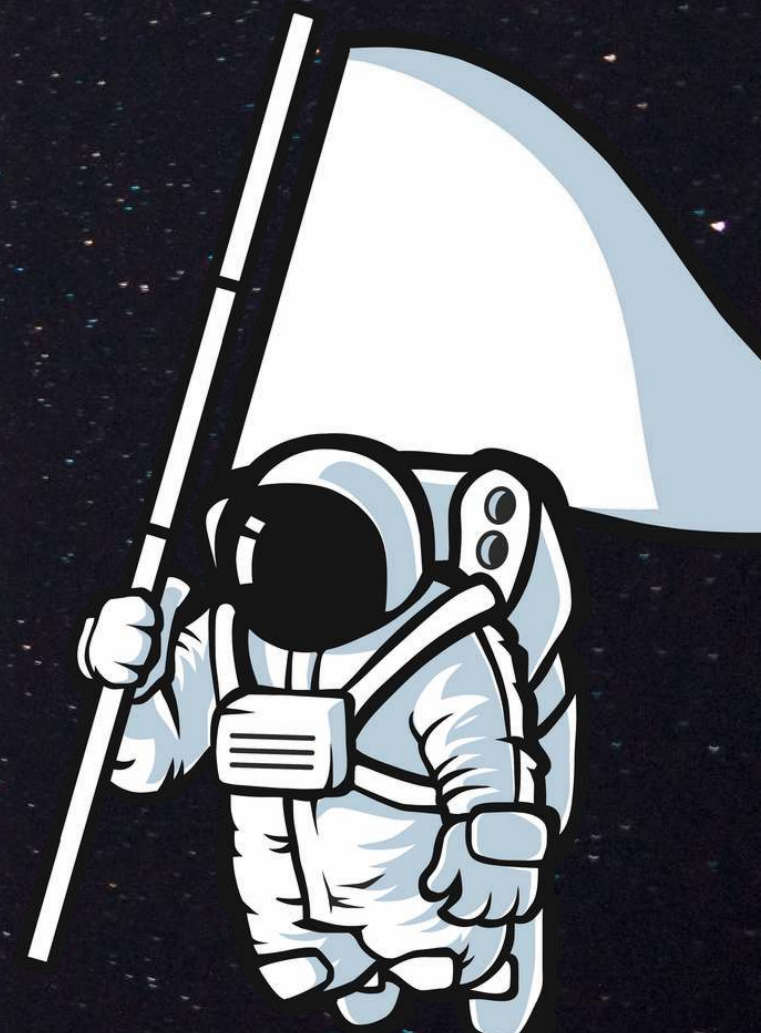
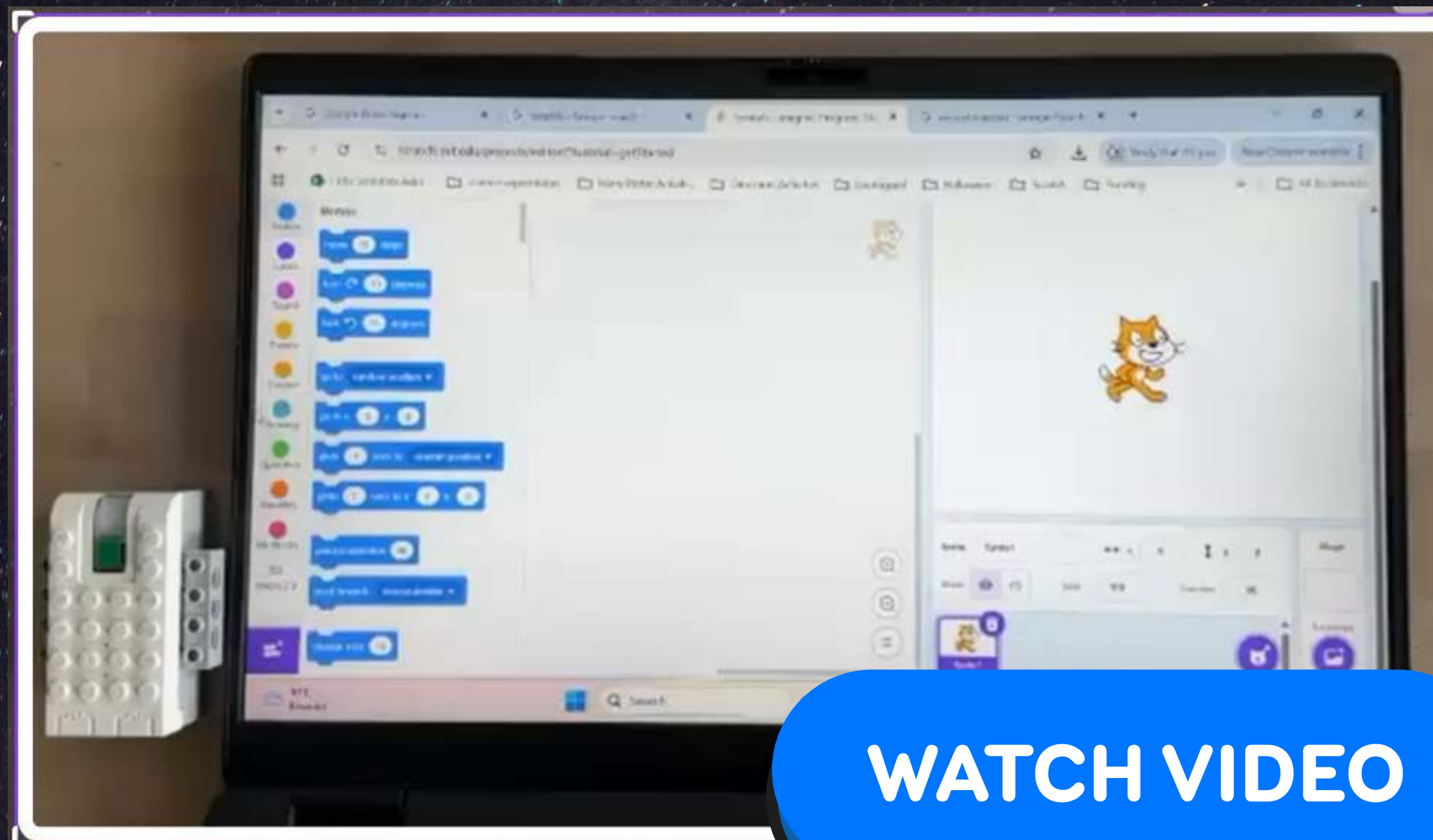
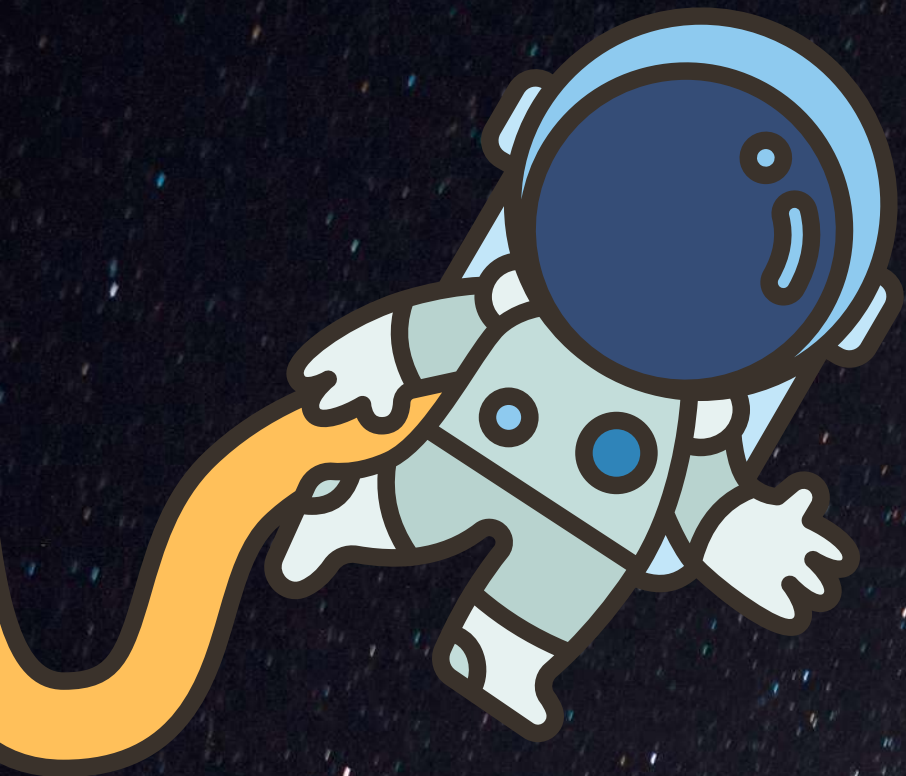
CODING 1

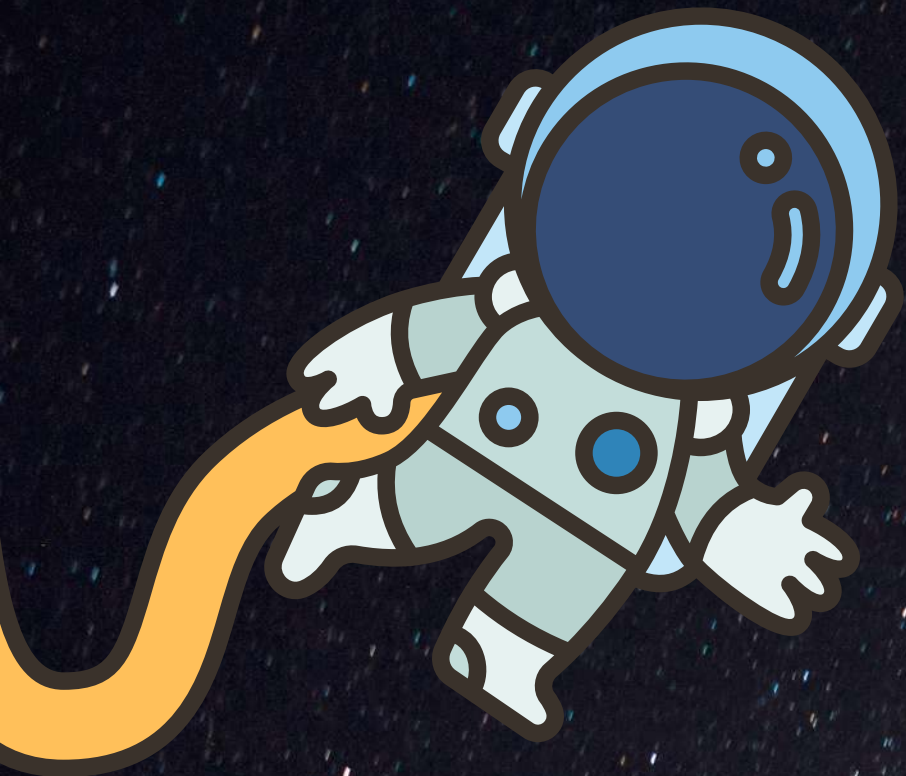
COLLECT THE STONES

SAMMLE DIE STEINE



CONNECT WITH SCRATCH



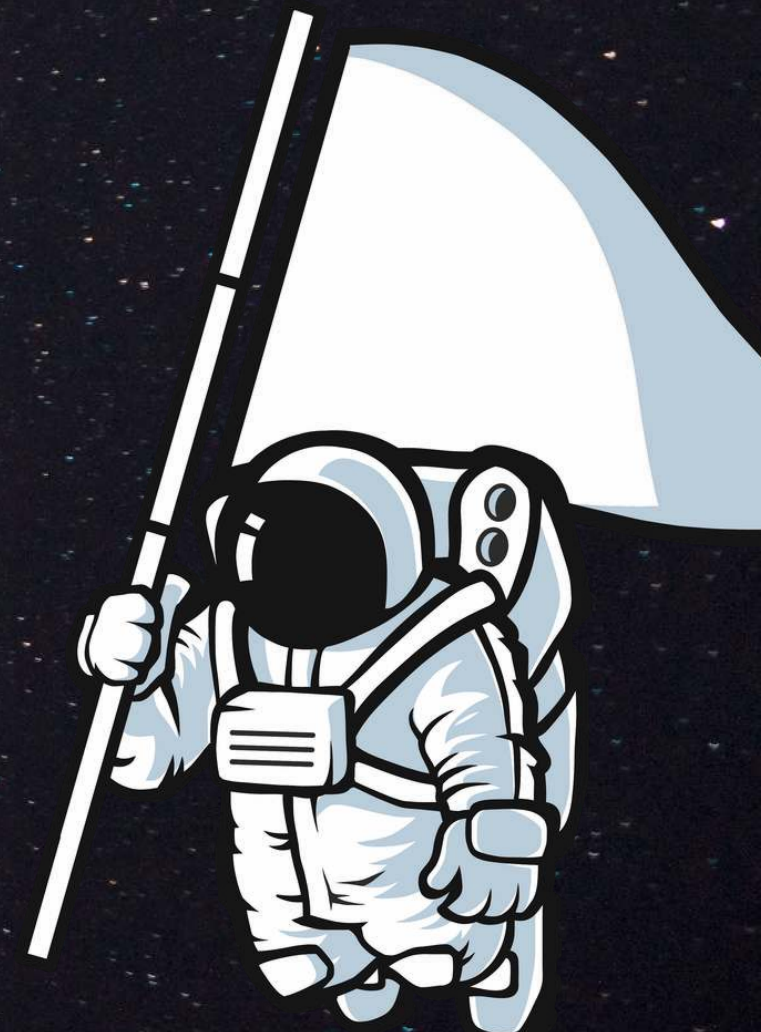


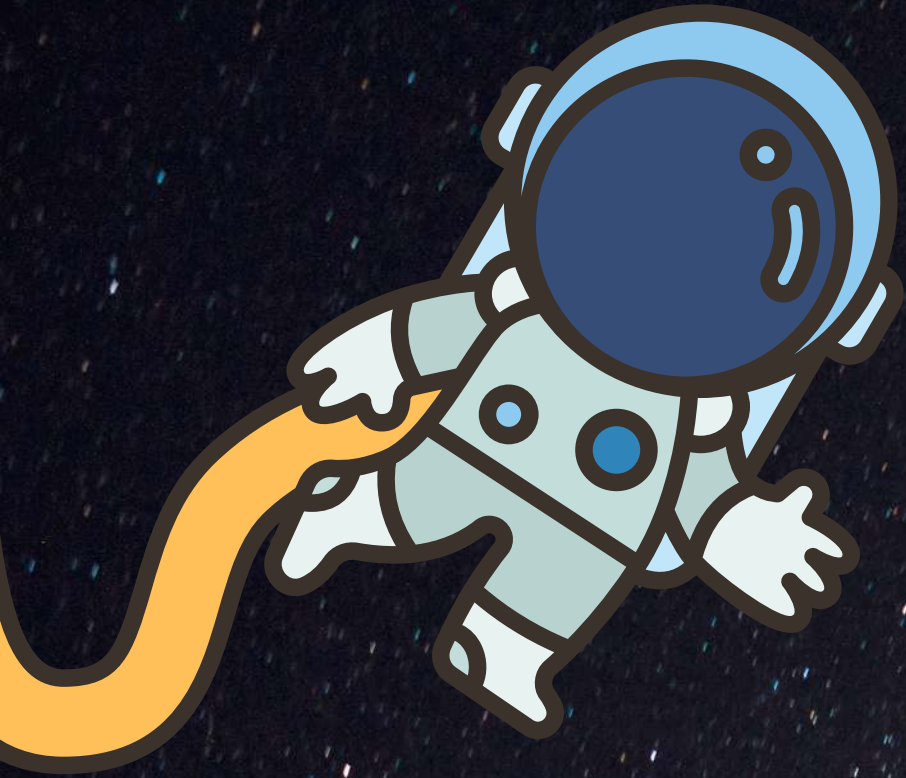
CODING 1

Collect the stones



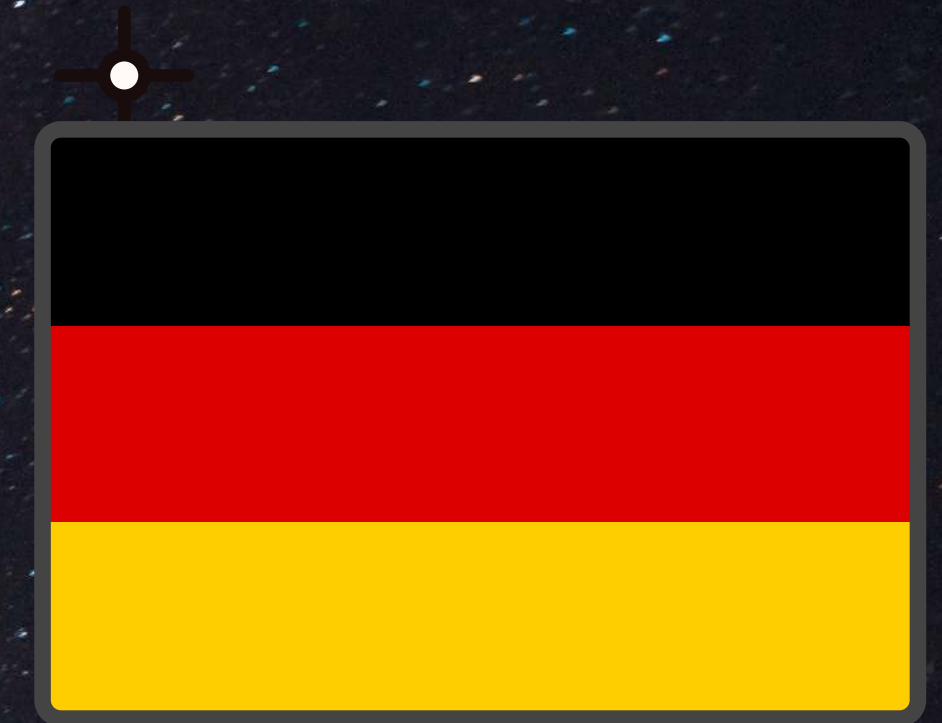
```
when clicked
  set motor power to 40
  set motor direction to this way
  turn motor on for 3 seconds
  set motor direction to that way
  turn motor on for 2 seconds
```





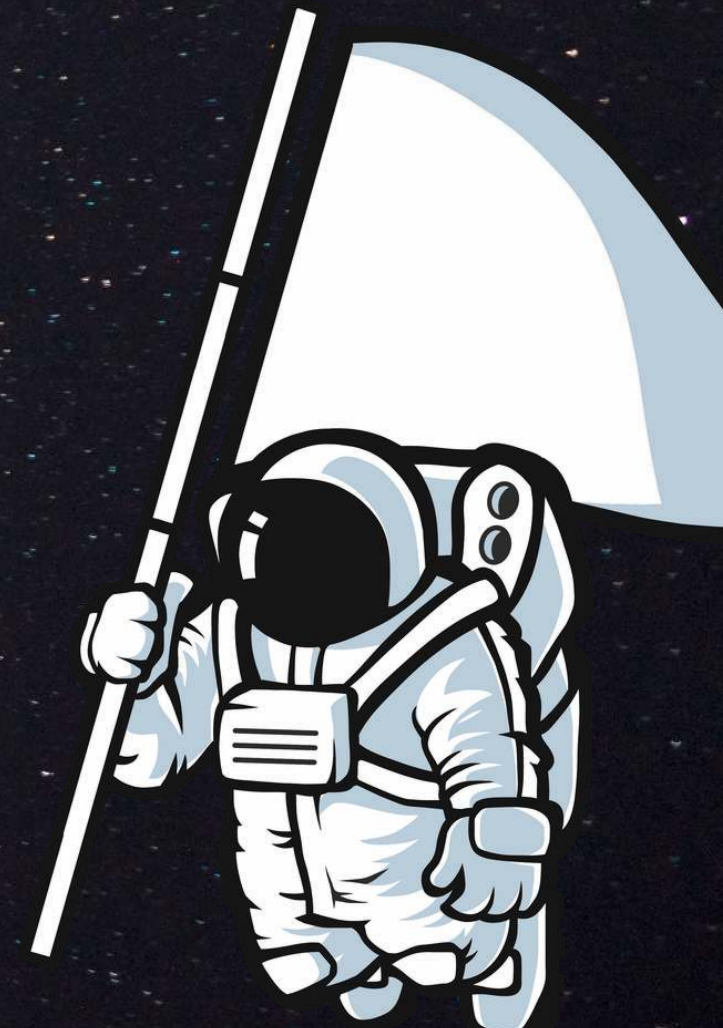
CODING 1

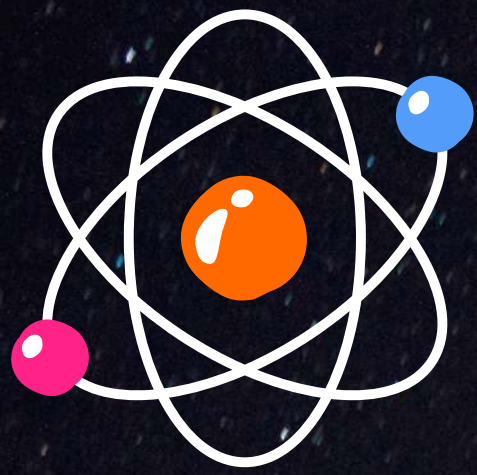
Sammele die Steine



Wenn  angeklickt wird

- setze Leistung von Motor ▼ auf 40
- setze Richtung von Motor ▼ auf rechtsherum ▼
- schalte Motor ▼ für 3 Sekunden an
- setze Richtung von Motor ▼ auf linksherum ▼
- schalte Motor ▼ für 2 Sekunden an

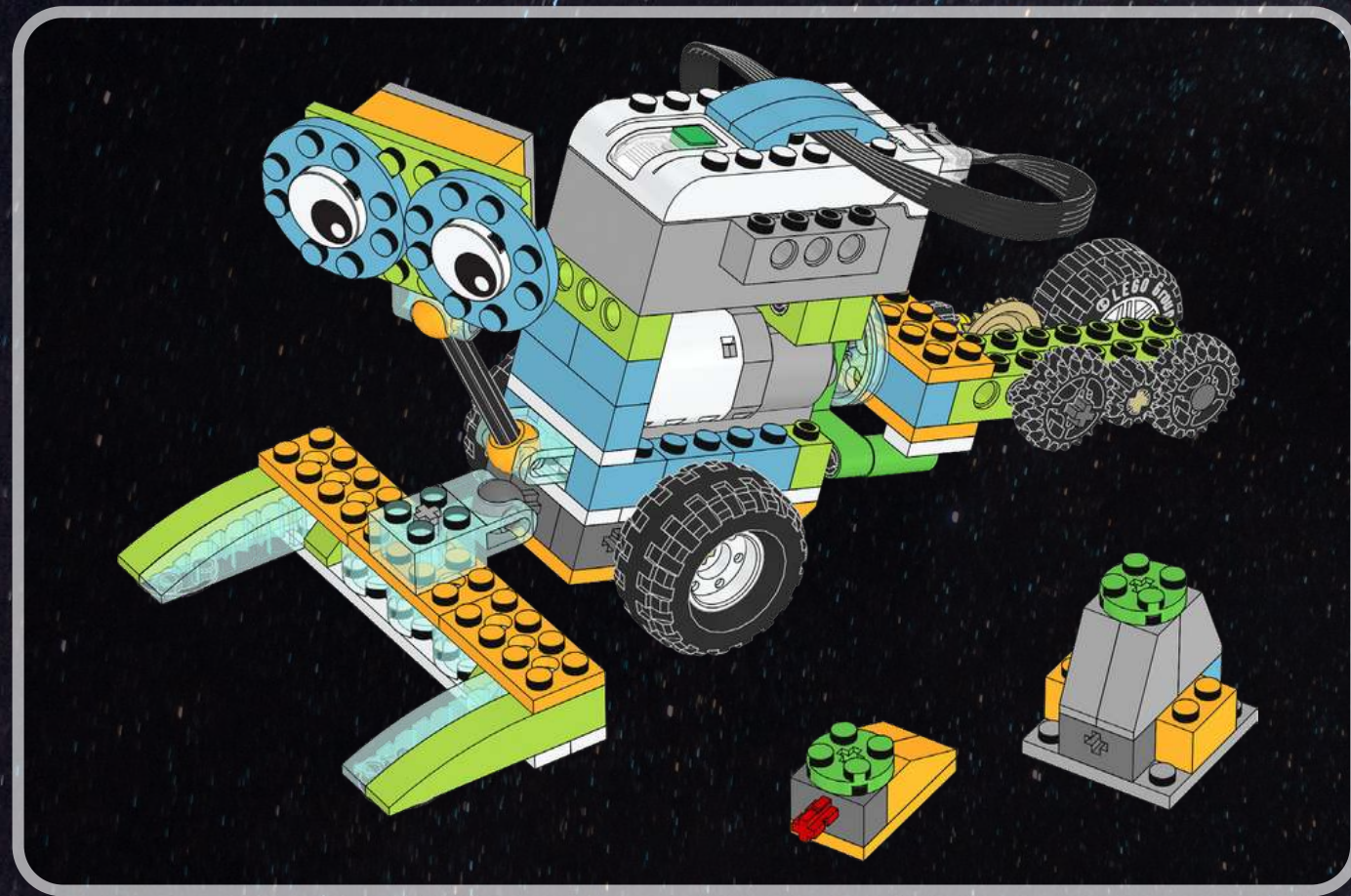




Challenge 1



GIVE YOUR ROVER A NAME AND LOGO
GIB DEINEM ROVER EINEN NAMEN UND EIN LOGO



NAME:

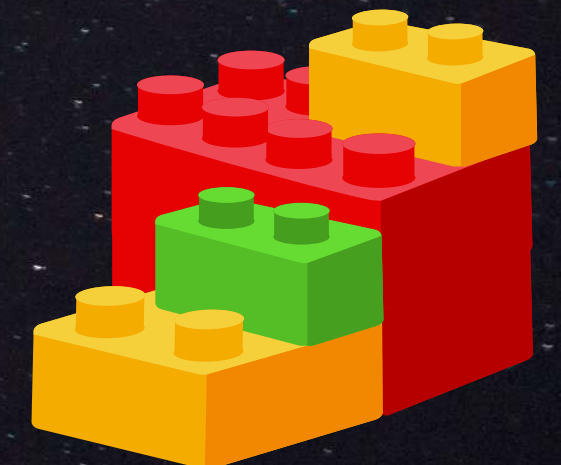
— . — . — . — . — . — .

Logo

Challenge 3

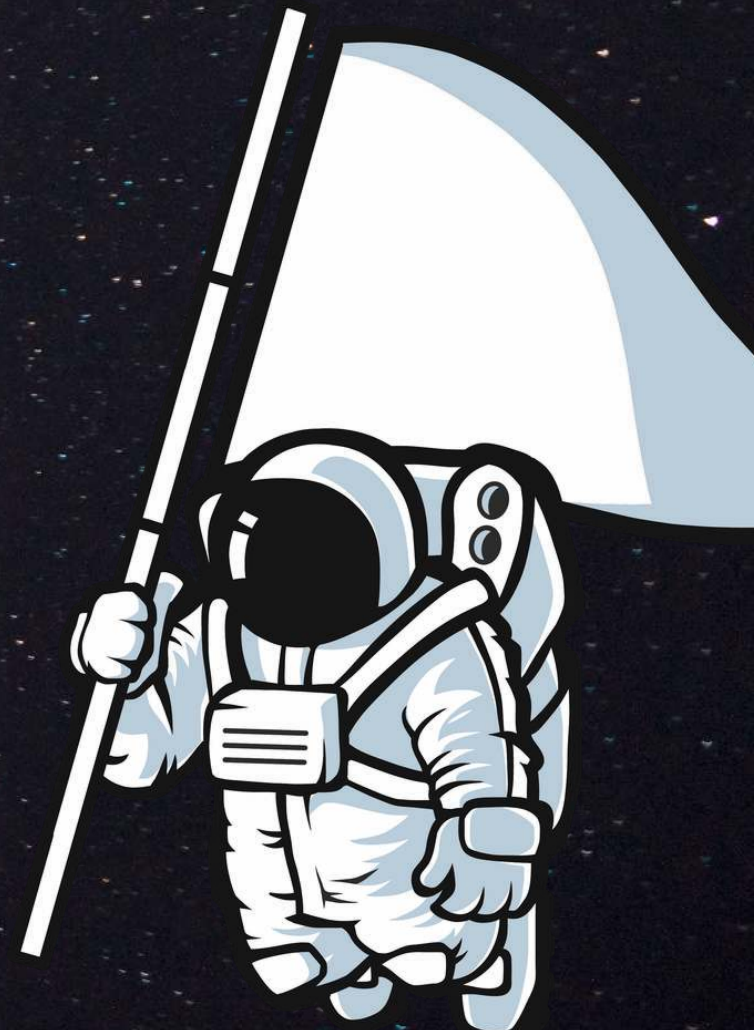
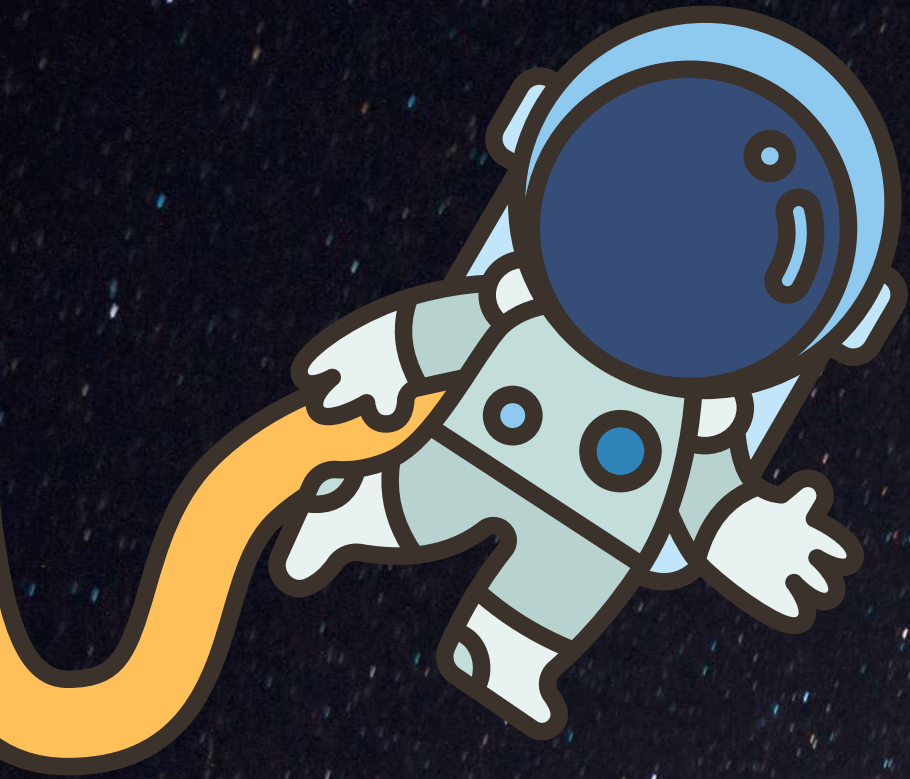
LUNAR RESEARCH BASE

Design your Lunar base research
Entwirf deine Mondbasis-Forschung



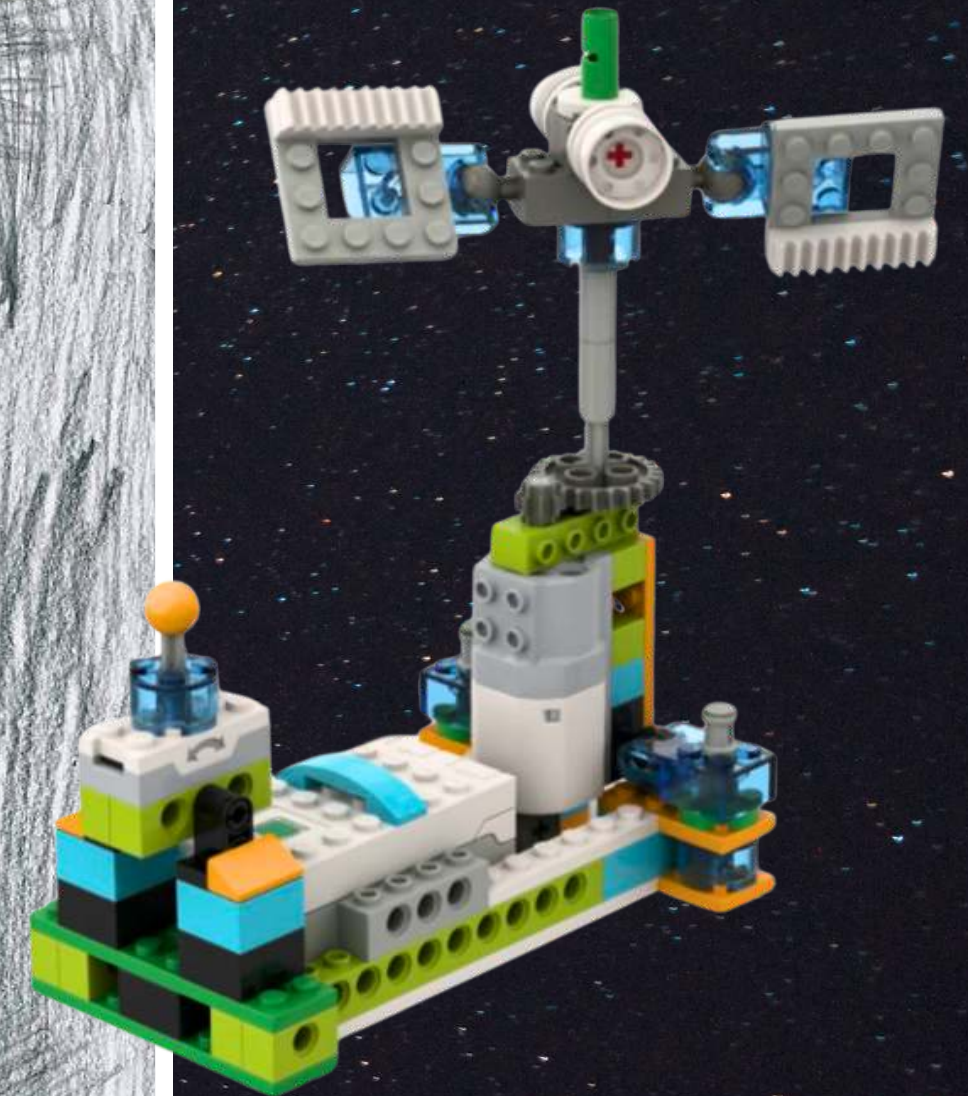


CODING 2



Challenge 3

MAKE A BACKGROUND FOR YOUR LEGO



Activities:

1. Art & Design (A in STEAM): Mission Aesthetics

This activity encourages creativity and visualization, directly supporting the lesson's existing planning (decomposition) and presentation (sharing) phases.

Activity Focus Description

Lunar Terrain Design Art/Set Design Challenge: students to transform their testing area into a detailed lunar landscape. Provide craft materials like large paper, aluminum foil, paint, or cardboard. Students can create:

- **Craters and Obstacles:** Small hills or bumps for the rover to navigate.
- **A Base Camp:** A designated, decorative area for the collected modules.
- **Star/Planet Backdrop:** A paper background to give the mission more context.
- **Rover Branding & Detail Art/Engineering Focus** on the visual design of the robot. Have students give their rover a name and logo. They can use non-electronic LEGO bricks, small paper flags, stickers, or markers to add non-structural aesthetic details. The design must not interfere with the robot's function.

Technical Blueprint Art/Planning Before students start building or programming, have them act as "Aerospace Architects." They must sketch a blueprint of their robot and a map of the mission trajectory, including the planned turns, distances, and module locations. This visual planning reinforces the decomposition (breaking the problem down) skill.

2. Expanded Science & Engineering (S & E in STEAM)

These activities extend the mission beyond just movement, integrating real-world scientific and engineering problems.

Activity Focus Description

Lunar Habitat Challenge Engineering/Science: After the robot collects the base modules, challenge the students to use extra LEGO bricks or simple materials (like toothpicks and marshmallows, or paper and tape) to design and build a small habitat for the future astronauts. The habitat must be engineered to protect against at least one lunar hazard, such as:

- Radiation (shielding)
- Micrometeorites (structural integrity)
- Extreme temperatures (insulation)

"Science Payload" Addition Technology/Engineering Challenge students to add a non-driving attachment to their robot that simulates a scientific instrument, like a sensor arm or a sample collector. This requires them to modify their existing design while maintaining the core functionality, emphasizing the iterative engineering process.

3. Math & Data (M in STEAM)

The core lesson already touches on the relationship between motor power and time. This activity formalizes that relationship.

Activity Focus Description

Distance and Power Graphing Math/Data Analysis Have students conduct an experiment with their robot:

- **Test:** Run the robot for a fixed amount of time (e.g., 5 seconds) using different Power Levels (e.g., 2, 4, 6, 8, 10).
- **Measure:** Use a ruler or measuring tape to record the distance traveled for each power level.
- **Graph:** Plot their findings on a simple bar graph or coordinate plane.
- **Predict:** Challenge them to use the data to predict exactly what Power Level they need to set to travel a specific, new distance (e.g., "How far do we go in 5 seconds at Power Level 7?").